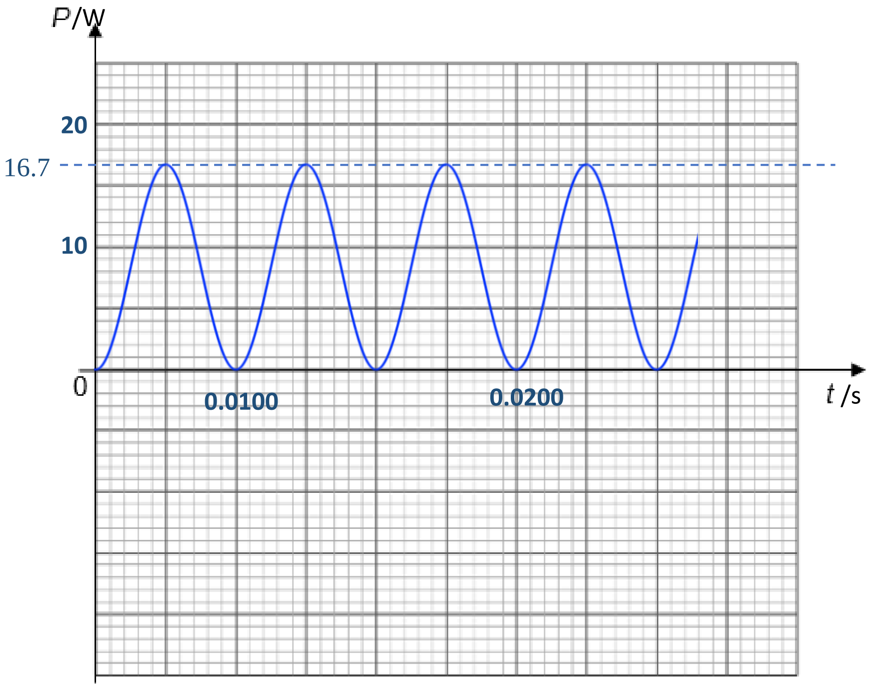
Q5		
(a)(i)	$\omega = \frac{2\pi}{T} = 628 \Rightarrow T = 1.00 \times 10^{-2}$ s	A1
(a)(ii)	$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{15}{\sqrt{2}} = 10.6$ V = 11 V	A1
(a)(iii)	$\frac{1}{R_{eff}} = \frac{1}{12.0} + \frac{1}{3.0+6.0} = \frac{7}{36} \Rightarrow R_{eff} = 5.1429\ \Omega$	M1
	$I_0 = \frac{V_0}{R} = \frac{15}{5.1429} = 2.92$ A = 2.9 A	A1
(a)(iv)	V_{rms} across 6.0 Ω resistor = $\frac{V_0}{\sqrt{2}} = \frac{(15 \times \frac{6.0}{9.0})}{\sqrt{2}} = \frac{10.0\text{ V}}{\sqrt{2}} = 7.071\text{ V}$	M1
	Mean power = $\frac{V^2}{R} = \frac{7.071^2}{6.0} = 8.33$ W = 8.3 W	A1
(b)	 $P_0 = 8.33 \times 2 = 16.7\ W$	
	B1 : Correct shape (sin^2 graph) *B1: at least 2 cycles shown with period labelled correctly ($t \geq 2T$ i.e. 4 maximum power in graph)* *B1: correct peak value labelled on graph.*	B3

Q6		